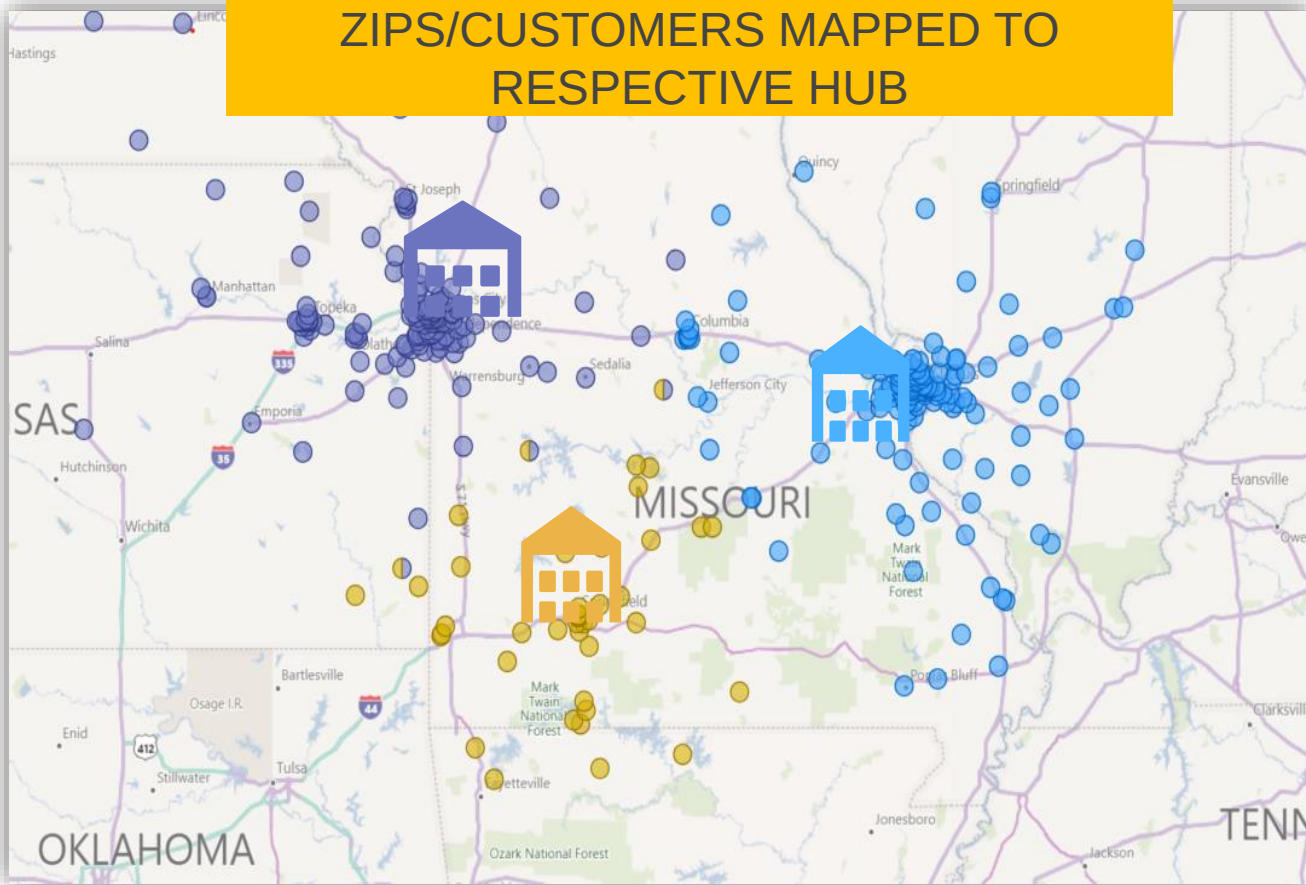


BACKGROUND

ZIPS/CUSTOMERS MAPPED TO RESPECTIVE HUB



- One important factor parts planned at various warehouses/hubs depend on:
 - **Active customers at zip code and their part requirement**
 - Customers at a particular zip code are all mapped to a hub within 130 miles for **same business day agreements (contract with logistic partner)**

OBJECTIVE SUMMARY

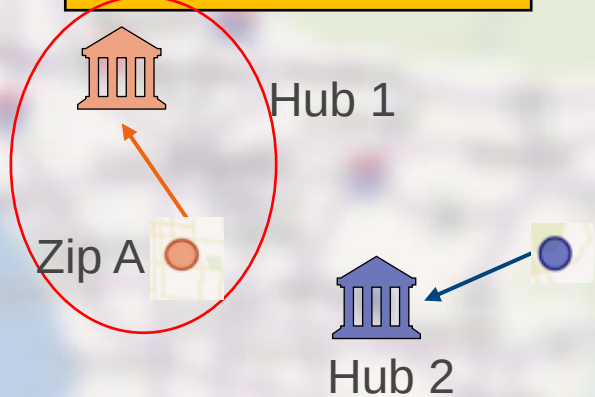
OBJECTIVE

Find opportunities to reduce costs on US hub network by taking calculated risks (e.g., increasing 130 miles radius for certain part/zip pairs)

- Updating assignment of zip/ SKU (part/zip) to hub based on distance, demand, inventory cost, and other costs.
- Closing or opening new hubs.

FORMULATION TYPE - INTEGER PROGRAM

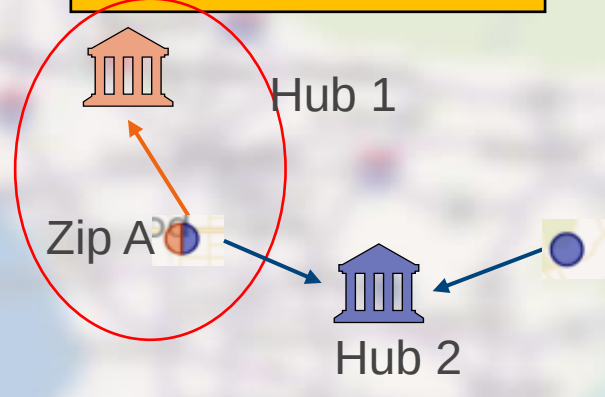
ZIP LEVEL



Entire Zip is mapped to a hub i.e. all customer part demand at a Zip satisfied by Zip's hub

Eg. Zip A for all parts mapped to Hub 1

SKU LEVEL



A SKU (part-zip) mapped to a hub i.e. customer at certain zip with certain part demand satisfied by SKU's hub

Eg. Zip A - Part A mapped to Hub 1;
Zip A - Part B mapped to Hub 2

COST STRUCTURE

- **Transportation** cost (From hub to customer)
- **Inventory** cost (Sum of unit price of the parts planned)
- **Storage** cost (Facility storage cost)
- **Transfer** cost (Move parts from one hub to another one)

INITIAL ASSUMPTIONS

- A zip / SKU (part/zip) can only be assigned to one hub
- Customers are located on the centroid of the zip code
- Hub located at the centroid of the zip code
- Max distance to reach within 4 hours is between 100-180 mi; Ground transportation only;
- All packages are under 50 Lbs.
- Demand volume will be similar than last 12 months
- Assumes 1 piece is required to plan

FORMULATION

FORMULATION ON SKU LEVEL (IP)

$k \in K$ (Part space) $j \in J$ (Hub space) ; $i \in I$ (Zip space)

Parameters

P_k – Part k price;

T_j - 1 if active facility/hub, 0 closed (binary)

C – Transfer cost;

h^s – Static Transportation cost for first 20 miles

h^d _ Dynamic Transportation cost per mile after 20

b_{ik} – No. of dispatches to zip i for part k ;

Q_{ik} -> 1 if part k needed for zip i else 0 (binary)

S_{lim}, S_{var} – Fixed storage cost/hubs upto limit L , Storage cost/part over limit L (qty.)

d_{ij} – Distance beyond 20 miles between zip i and hub j

d_s - 110 (for customer experience penalty when going beyond 130 miles)

$\lambda_1, \lambda_2, \lambda_3$ – Weightage (**Recommendation:** trial and error)

Decision Variables

Z_{ijk} : 1 when zip code i , part k supported by hub j else 0 (binary) (**Recommendation:** Preprocessing step to cap at 180 miles)

Y_{jk} : 1 part k assigned to hub j else 0 (binary)

X_j : 1 when hub j active else 0 (binary)

Resultant variable

W_j : count of parts planned over hub storage limit L

FORMULATION ON SKU LEVEL (IP) contd.

Objective:

Minimize inventory, storage cost, transfer cost and transportation costs

$$\sum_j \sum_k Y_{jk} * P_k + \sum_j X_j * S^{limit} + \sum_j W_j * S^{var} + \sum_j \sum_k Y_{jk} * (1 - T_j) * C + \lambda_1 * \sum_i \sum_j \sum_k h^s * b_{ik} * Z_{ijk} + \lambda_2 * \sum_i \sum_j \sum_k h^d * b_{ik} * d_{ij} * Z_{ijk} + \lambda_3 * \sum_i \sum_j \sum_k b_{ik} * (d_{ij} - d_s) * Z_{ijk}$$

Constraints

- Every zip code,part (SKU) assigned to one hub location

$$\sum_j Z_{ijk} = 1$$

- If part k and zip i assigned hub j , location must be open

$$\sum_i \sum_k Z_{ijk} Q_{ik} \leq Y_{jk}$$

- To get active hub Variable X_j (1 if j active, 0 if not)

$$\sum_k Y_{jk} * \frac{1}{B} \leq X_j$$

Where B is the upper bound of parts planned (max. number of parts planned)

- To get count of parts over storage limit (L) to charge storage variable cost

$$\sum_k Y_{jk} - L \leq W_j$$

$$W_j \geq 0$$